

The Human Eye and the Colourful World

How the human eye forms images, why vision defects occur and how they are corrected, and how prism, rainbow and scattering create the colours around us.

Structure of the Human Eye

The human eye is a natural optical instrument. It forms a real and inverted image on the retina. The brain then interprets this image as erect.

Main parts of the eye are:

- **Cornea.** The transparent bulge on the front surface. Most of the refraction of light entering the eye happens here. It is made of transparent tissue.
- **Iris.** The coloured circular diaphragm behind the cornea. It controls the size of the pupil.
- **Pupil.** The black opening in the centre of the iris. It controls the amount of light that enters the eye. It becomes small in bright light and large in dim light.
- **Eye lens.** A convex lens made of transparent and flexible material. It is held by ciliary muscles.
- **Ciliary muscles.** They change the curvature of the eye lens. This changes the focal length of the lens.
- **Retina.** The light sensitive screen at the back of the eye. It has a large number of light sensitive cells. These are rods and cones. The image is formed on the retina.
- **Optic nerve.** It carries electrical signals from the retina to the brain.
- **Blind spot.** The point where the optic nerve leaves the retina. There are no light sensitive cells here. So an image formed on this spot is not visible.

Working: Light enters through the cornea, passes through the pupil, the eye lens and the vitreous humour, and forms an image on the retina. The retina converts the light into electrical signals. These signals go to the brain through the optic nerve, and the brain gives the sense of sight.

Key distances

Least distance of distinct vision (Near point): The smallest distance at which a normal eye can see an object clearly without strain. It is 25 cm for a young adult. This is denoted by D .

Far point: The farthest distance up to which a normal eye can see clearly. For a normal eye it is infinity.

Power of Accommodation

The ability of the eye lens to adjust its focal length is called power of accommodation.

The ciliary muscles change the curvature of the eye lens. When we look at a distant object, the ciliary muscles relax, the lens becomes thin and its focal length increases. When we look at a near object, the ciliary muscles contract, the lens becomes thick and its focal length decreases.

Due to this adjustment, a normal eye can focus objects from its near point to its far point on the retina. The eye lens has its greatest optical power when focusing on a nearby object. The range through which the eye can adjust decreases with age.

Power of accommodation

The ability of the eye lens to change its focal length by changing its curvature with the help of ciliary muscles, so that objects at different distances can be seen clearly.

Defects of Vision and Correction

A normal eye can see objects from 25 cm to infinity. Sometimes the eye loses its power of accommodation and cannot see clearly in some range. This is called a defect of vision. The three common defects are myopia, hypermetropia and presbyopia. Cataract is also a defect due to old age.

Myopia (Near sightedness)

A person with myopia can see near objects clearly but cannot see distant objects clearly. The far point moves nearer than infinity.

Cause: (i) Excessive curvature of the eye lens, so focal length becomes too small. (ii) Elongation of the eyeball, so the distance between lens and retina increases. In both cases, the image of a distant object is formed in front of the retina instead of on the retina.

Correction: A concave lens of suitable power is used. A concave lens diverges the rays. So the image is shifted back onto the retina.

Worked Example

A myopic person uses spectacles of power -2.0 D. What is the focal length of the lens?

$$P = 1 / f \text{ (in metres)} \quad \text{so} \quad f = 1 / P = 1 / (-2.0) = -0.50 \text{ m} = -50 \text{ cm}$$

The negative sign shows it is a concave lens. Focal length is 50 cm.

Hypermetropia (Far sightedness)

A person with hypermetropia can see distant objects clearly but cannot see near objects clearly. The near point moves away from 25 cm.

Cause: (i) The focal length of the eye lens is too large. (ii) The eyeball has become too small. So the image of a near object is formed behind the retina.

Correction: A convex lens of suitable power is used. A convex lens converges the rays, so the image is formed on the retina.

Worked Example

A hypermetropic eye has a near point at 75 cm. What power of convex lens is needed to see clearly at 25 cm? Here, the lens must form the image of an object at 25 cm at a distance of 75 cm, so that the eye can see it.

$$u = -25 \text{ cm}, \quad v = -75 \text{ cm}$$

$$1/f = 1/v - 1/u = 1/(-75) - 1/(-25) = -1/75 + 1/25 = 2/75$$

$$f = 75 / 2 = 37.5 \text{ cm} = 0.375 \text{ m} \quad \text{so} \quad P = 1 / 0.375 = +2.67 \text{ D}$$

Presbyopia

Presbyopia occurs in old age. The power of accommodation of the eye decreases due to weakening of ciliary muscles and loss of flexibility of the eye lens. A person cannot see near objects clearly.

It can be similar to hypermetropia. It is corrected by convex lenses. Often a person needs bifocal lenses, with a convex lens for near vision and a concave lens for distant vision in the upper part.

Cataract

In cataract, the crystalline lens becomes cloudy or milky. This reduces vision. Cataract can be treated surgically; the cloudy lens may be replaced with an artificial lens.

Refraction through a Prism

A glass prism has two triangular bases and three rectangular sides. These sides are inclined to each other. The angle between two lateral faces is called the angle of the prism.

When a ray of light passes through a prism, it bends twice.

- At the first surface, the ray bends towards the normal, because it goes from air to glass.
- At the second surface, the ray bends away from the normal, because it goes from glass to air.

The ray inside the prism bends towards the base of the prism. The emergent ray is deviated by an angle from its original direction. This angle is called the angle of deviation, denoted by D .

The peak of the prism is called the apex. The side opposite to the apex is the base.

Angle of deviation

The angle between the direction of the incident ray and the direction of the emergent ray after refraction through a prism.

Dispersion and Rainbow

White light is made of seven colours. When white light passes through a prism, it splits into its seven colours. This splitting is called dispersion.

The seven colours are Violet, Indigo, Blue, Green, Yellow, Orange and Red (VIBGYOR). Violet bends the most and red bends the least. This is because the refractive index of glass is slightly different for different colours. It is highest for violet and lowest for red, and wavelength of red is largest.

Recombination: Newton showed that if the dispersed light is passed through a second identical prism placed in inverted position, the seven colours recombine to give white light again. This proves that white light is a mixture of seven colours, and the prism itself does not produce colours.

Rainbow

A rainbow is a natural example of dispersion. It is formed after rain, when small water droplets in the air act like tiny prisms.

Steps: (i) Sunlight enters a water droplet and refracts. (ii) It disperses into seven colours. (iii) It reflects from the back surface of the droplet (internal reflection). (iv) It refracts again while coming out of the droplet. So the observer sees a rainbow opposite to the sun. The inner arc is violet and the outer arc is red.

Scattering of Light

When light falls on very small particles, it is scattered in different directions. The scattering depends on the size of the particle and the wavelength of light. Shorter wavelengths are scattered more than longer wavelengths. This is called scattering of light.

Blue light has a shorter wavelength, so it is scattered more than red light.

Tyndall Effect

When a beam of light passes through a colloidal solution or through smoke or fine dust, its path becomes visible because the colloidal particles scatter the light. This is called the Tyndall effect. Examples are a beam of light in a dark room with dust, and sunlight through the canopy of a forest.

Why is the sky blue

The fine particles in the atmosphere scatter blue light more than red light. When sunlight passes through the atmosphere, blue light is scattered in all directions. When we look up, this scattered blue light reaches our eyes. So the sky appears blue.

Danger signals are red

Red light has the longest wavelength and is scattered least. So it can travel far and is visible even through fog and smoke.

Atmospheric Refraction

The atmosphere is not uniform. Its density changes with height. Light coming from stars or the sun bends when it enters the atmosphere. This is atmospheric refraction.

- **Twinkling of stars:** Stars are point sources of light. Their light passes through layers of air with different density. The path keeps changing slightly. So the apparent position of the star shifts and its brightness changes. This looks like twinkling. Planets are nearer and appear as a collection of many point sources, so the effect averages out and planets do not twinkle.
- **Stars appear higher than they actually are:** Due to refraction, a star appears slightly higher in the sky than its real position when seen near the horizon.
- **Advance sunrise and delayed sunset:** The sun appears about two minutes before actual sunrise and two minutes after actual sunset. This is because the rays from the sun below the horizon are bent by the atmosphere and reach the observer.

Summary

- Cornea does most of the refraction. Iris and pupil control light. Eye lens forms image on retina. Retina sends signals to brain through optic nerve.
- Power of accommodation is the ability of the eye lens to change its focal length with ciliary muscles.
- Least distance of distinct vision is 25 cm. Far point is infinity for a normal eye.
- Myopia: cannot see distant objects, corrected by concave lens. Hypermetropia: cannot see near objects, corrected by convex lens. Presbyopia: loss of

accommodation in old age, corrected by bifocal lens. Cataract: cloudy lens, corrected by surgery.

- A prism deviates light towards its base. Angle of deviation is between incident and emergent ray.
- White light splits into seven colours through a prism. This is dispersion. Violet deviates most, red least. Water droplets form a rainbow by dispersion, internal reflection and refraction.
- Scattering: shorter wavelengths are scattered more strongly. This explains the blue sky, Tyndall effect and use of red for danger signals.
- Atmospheric refraction causes twinkling of stars and advance sunrise and delayed sunset.

Practice Questions

1. Draw a labelled diagram of the human eye and explain the function of cornea, iris, pupil, eye lens and retina.

2. What is power of accommodation? How do ciliary muscles help in viewing objects at different distances?

3. A person cannot see objects beyond 1.2 m clearly. Name the defect and state the type and power of lens needed if his far point should be infinity. Calculate focal length.

4. Explain why a prism splits white light into seven colours. Why does violet bend more than red?

5. What is the cause of a rainbow? Describe the events inside a water droplet that lead to a rainbow.

6. Why does the sky appear blue? Why are danger signals generally red?